

Experiment No(3) Sublimation.

* introduction:-

Some solids when heated pass directly into Vapour state without melting to liquid and by suitable arrangement.

The Vapour can be condensed the original solid in purer condition, this process is known as **Sublimation**.

* Apparatus:-

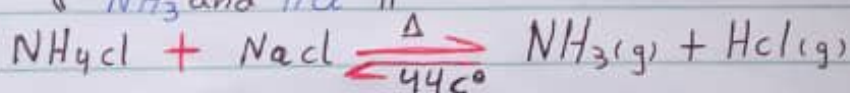
- | | |
|----------------|--------------|
| 1- thermometer | 2- crucible |
| 3- stand | 4- gas flame |
| 5- funnel | |

* chemicals:-

- | | |
|---|-----------------------------|
| 1- 1g(NH ₄ Cl _(s)) | 2- 1g(NaCl _(s)) |
|---|-----------------------------|

* procedure:-

- 1) balance before experiment, the crucible is clean and take its weight.
- 2) take 1g(NH₄Cl) and 1g(NaCl).
- 3) mix in crucible.
- 4) put the crucible on the gas flame or the stand.
- 5) cover the crucible by funnel
- 6) start heating up to 44°C.
- 7) During heating the gas flame to get two gases
« NH₃ and HCl »



Note:- NaCl_(s) $\rightleftharpoons$ NaCl_(s) « but remain of crucible ».

- 8) Balance after heating the crucible with the solids and take its weight.

★ Result :-

- 1) Weight of crucible before heating (w_1) ?
- 2) Weight of crucible before heating with solids (w_2) ?
- 3) Weight of crucible with the solids after heating (w_3) ?

★ Aim of experiment :-

- to understand the concept of sublimation.
- and observe how sublimation helps in purifying solids.
- and to identify substances that undergo sublimation. Like
the:- NH_4Cl